

# $\pi$ Tantum

*a.k.a.* Tempus Tantum

school timetable generation system

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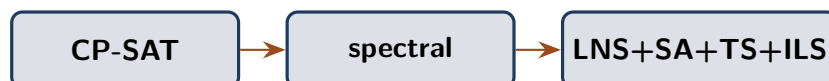
*“Omnia, Lucili, aliena sunt, tempus tantum nostrum est.”*  
Lucius Annaeus Seneca, *Epistulae morales ad Lucilium*, I, 1

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## Manual

*A method for Italian-school timetabling*  
(English edition)

From an idea by Fernando Gargiulo, Giovanni Borghi,  
Matteo Mariani, Stefano Bertozzi



Giovanni Borghi

May 7, 2026

Repository: <https://github.com/gborghi/timetable>  
Documentation source: docs/

## Colophon

This English edition is set in EB Garamond for body text and Lato for headings, with Inconsolata for code blocks. Compilation is done with lualatex (TeX Live), biber for bibliography and makeindex for the analytical index. The L<sup>A</sup>T<sub>E</sub>X source lives in docs/manual\_en.tex with chapters in docs/manual/chapters\_en/; the bibliography is shared with the Italian edition in docs/manual/bibliography.bib. The build pipeline is in docs/build\_manual.sh which emits **both** manual.pdf (Italian) and manual\_en.pdf (English) in a single invocation.

**Note on translation:** the Italian edition remains the primary reference. This English edition fully translates the introduction, the BP-related chapters and the benchmark chapter; all other chapters carry an English summary that points the reader to the corresponding Italian chapter for the full narrative. Where Italian-school context is necessary (for instance the legal frame of the *contratto collettivo nazionale*, the role of the *insegnante di sostegno*, or the meaning of *indirizzo di studio*), explanatory notes are added in the English text.

The repository version corresponds to the most recent commit visible in `git log --oneline -- docs/`. Errors and suggestions can be reported as issues on the GitHub repository at <https://github.com/gborghi/timetable>.

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# Introduction

“Omnia, Lucili, aliena sunt, tempus tantum nostrum est.”

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Lucius Annaeus Seneca, *Epistulae morales ad Lucilium*, I, 1

$\pi$ Tantum is a web application that assists in the composition of Italian-school timetables. It originates from ideas discussed and developed by Fernando Gargiulo, Giovanni Borghi, Matteo Mariani and Stefano Bertozzi.

The school timetabling problem consists of assigning, to each *cattedra* – the pair formed by a teacher and a class for a specific subject – a set of weekly hours distributed over days and time slots, such that all the following constraints are honoured: teacher availability, curricular hour coverage, classroom and equipment compatibility, and the constraints that derive from the Italian national collective contract for the school sector. This is a combinatorial problem classified as NP-hard, which means that as a school grows in size there is no fast algorithm capable of finding an exact optimal solution.

$\pi$ Tantum delegates the actual solving to the CP-SAT solver in Google OR-Tools, chosen for its maturity in constraint programming applied to scheduling problems. To scale beyond the size that the solver would handle directly, the system decomposes the problem into smaller sub-problems via four alternative strategies: spectral decomposition of the bipartite class-teacher graph, temporal decomposition by day of the week, curriculum-based decomposition, and balanced  $k$ -way METIS partitioning. On top of these, a cascade of metaheuristics (Large Neighborhood Search, Adaptive LNS, Simulated Annealing, Tabu Search, Iterated Local Search and Variable Neighborhood Search) improves the quality of the solution found.

The application backend is written in Python on FastAPI; data persistence is delegated to SQLite. The user interface is based on SvelteKit (in Italian for the production deployment), organised into sixteen tabs that cover the whole lifecycle of a timetable: registry import, constraint configuration, generation, manual editing with real-time feasibility checks, and day-to-day handling of absences and substitutions.

The following chapters describe the problem formally, the entities involved, the user interface, and each of the algorithms that  $\pi$ Tantum employs to compute the timetable. The exposition is intentionally multi-layered: a non-technical reader can follow the narrative thread without engaging the mathematical formulas, while a developer will find the implementation details, code references, and source-file pointers necessary to extend the system.

This English edition is a parallel translation of the Italian manual (docs/manual.tex). The Italian edition remains the primary reference; this English edition is intended for international readers and contributors. Where Italian-school context is necessary (e.g. the legal frame of the *contratto collettivo nazionale*, the role of *insegnante di sostegno*, or the meaning of *indirizzo di studio*), explanatory notes are added.

# 1 The timetabling problem

Formal statement of the school timetabling problem: cattedre as (teacher, class, subject) triples; weekly hour distribution across days and time slots; HARD constraints (no class/teacher overlap, hour coverage, consecutive hours starting at 8 with presence at  $h=11$ ) and SOFT objectives (daily uniformity, free-day preferences). NP-hardness and the rationale for decomposition + metaheuristics.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/problema\_\_timetabling.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 2 piTantum overview

Three-tier architecture: engine (CP-SAT solver core), backend (FastAPI + SQLite), frontend (SvelteKit). Pipelines exposed as async runs via `/api/optimize/*`. The 16 UI tabs covering registry, constraints, generation, manual editing, daily operations.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/panoramica\_\_pitantum.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

### 3 The timetable as a structural object

The Lesson dict (teacher, class, subject, day, hour): 1 as the canonical representation; round-trip with the Solution table; the role of dc\_value in encoding Phase A's day-count distribution.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/orario\_oggetto\_strutturale.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 4 Installation

Python 3.13 + ortools + scipy + matplotlib + FastAPI + SQLAlchemy + Alembic; SvelteKit for the frontend. Cypress + Playwright for E2E. docker-compose.test.yml orchestrates the test stack.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/installazione.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 5 Constraints (HARD + SOFT)

HARD: no overlap (teacher in one class at a time, class with one teacher per slot), hour coverage of all cattedre, daily start at h=8, consecutive hours, presence at h=11 (the H<sub>1</sub>/H<sub>2</sub>/H<sub>3</sub> trio). Native locks. C<sub>1</sub> (compresenza/coteach), C<sub>2</sub> (sostegno DVA), C<sub>3</sub> (potenziamento, parallel intra-class, group inter-class). SOFT: daily uniformity, free-day preferences, sixth-hour penalties, five/one day load.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/vincoli.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

### 5.1 Plessi: multiple sites in the same school

In Italy many secondary schools are spread across multiple physical sites – the *Sede Centrale* (main building), one or more *succursali* (branch buildings, e.g. *Succursale Via Garibaldi*), or a dedicated *plesso* for a specific track (e.g. *Plesso ITT* for the technical track). Moving teachers or classes between sites during the day costs real time and introduces constraints that cannot be modelled with H<sub>1</sub>/H<sub>2</sub>/H<sub>3</sub> alone.

The **Plessi** module (tab /plessi) handles:

- **site registry** (name, code, address);
- **classroom** → **plesso** association (each classroom belongs to exactly one site);
- two kinds of behavioural rules: **commuting rules** (minimum gap to move between two sites) and **entity policies** (global constraints on how many sites a teacher or class may attend).

#### 5.1.1 Commuting rules: the gap between two sites

A commuting rule applies to an ordered pair (from\_plesso, to\_plesso) and specifies the minimum number of hours that must separate two consecutive lessons of the same entity when the classroom changes site. Relevant fields:

- from\_plesso\_id, to\_plesso\_id: the site pair.
- entity\_kind: who is affected (teacher, class, group).
- entity\_id: optional – if NULL the rule is *kind-wide* (applies to every teacher / class / group); if set it applies only to that specific entity.
- min\_gap\_hours: minimum number of empty hours required between the last lesson at site A and the first lesson at site B.
- symmetric: if true the rule applies in the opposite direction (B→A) as well.
- priority: tie-breaker when several rules overlap.

Typical Italian example: between *Sede Centrale* and *Succursale Via Garibaldi* a 60-minute travel time is required. The matching rule is (Centrale, Garibaldi, kind=teacher, entity\_id=NULL, min\_gap\_hours=1, symmetric=true): no teacher may have a lesson at Garibaldi at h=9 if they have one at Centrale at h=8.

Rule resolution always applies *specific beats generic*: if a rule with entity\_id=42 exists for teacher Rossi, it overrides the kind-wide rule. Among rules of equal specificity, higher priority wins.

### 5.1.2 Entity policies: per-person global constraints

An entity policy is a *global* constraint (not tied to a specific site pair) on the behaviour of a teacher or class. Three policies are supported:

- any – no global constraint, only commuting rules apply. This is the default.
- single\_plesso\_per\_day – on any given day the entity may have lessons in at most one site. Stricter than commuting rules: even with a 1-hour gap, switching sites mid-day is forbidden.
- single\_plesso\_total – the entity attends *only* site plesso\_id for the entire week. Every classroom assigned to that entity must belong to that site.

Concrete Italian examples:

- *Teacher Rossi does not want mid-day site switches*: the office creates a policy (entity\_kind=teacher, entity\_id=Rossi, policy=single\_plesso\_per\_day). On Monday all of Rossi's hours are at Centrale; on Tuesday all at Garibaldi; never a mixed day.
- *Class 3A (technical track) attends only Plesso ITT*: policy (entity\_kind=class, entity\_id=3A, policy=single\_plesso\_total, plesso\_id=ITT). Every classroom that hosts 3A lessons must belong to Plesso ITT.
- *All teachers, by default, avoid mid-day site switches*: policy (entity\_kind=teacher, entity\_id=NULL, policy=single\_plesso\_per\_day) – applies to every teacher except those with a more specific override.

Entity policies follow the same *specific beats generic* rule (per-entity override), then priority.

### 5.1.3 Pre-run validation

Before launching the optimisation the system runs a battery of checks on the plessi data (validate\_plessi\_rules). If anything is inconsistent the run is rejected with a list of violations. What is checked:

- empty or duplicate plesso code;
- commuting rule referencing a non-existent plesso;
- commuting rule with negative min\_gap\_hours or larger than the day length;
- duplicate kind-wide commuting rule (same site pair, same kind, NULL entity\_id, defined twice);
- single\_plesso\_total entity policy with no plesso\_id or with a non-existent site;
- entity policy with an unknown policy value;
- override for a non-existent entity (deleted teacher or class).

The GET /api/plessi/validate endpoint lets the UI display a live “plessi config OK / N issues” badge.

### 5.1.4 CP-SAT implementation

At solver level, commuting rules become *pairwise* constraints on the classroom-assignment variables:

$$\forall (h_a, h_b) \text{ adjacent such that } \text{room}(t, d, h_a) \in P_A, \text{room}(t, d, h_b) \in P_B : h_b - h_a \geq \text{min\_gap}_{P_A \rightarrow P_B}.$$

Entity policies become counter constraints: `single_plesso_per_day` is  $\sum_P \mathbf{1}[\exists h : \text{room}(t, d, h) \in P] \leq 1$  for each day  $d$ ; `single_plesso_total` fixes the domain of the classroom variables to the target site only.

All such constraints are *hard* by default. The roadmap envisages a *soft* variant with penalty for sporadic emergency overrides (e.g. last-minute substitute teaching), but the current version treats every plesso constraint as HARD.

### 5.1.5 Plessi via DSL (Step 3a-3c)

Starting in Step 3a, `PlessoCommutingRule` rows are fully expressible as DSL clauses thanks to two language additions (cf. cap. ??):

- the `l.classroom.plesso` chained path that, given a lesson, resolves the assigned room and then the plesso the room belongs to;
- the `consecutive(l1.slot, l2.slot)` predicate, true when the two lessons fall on the same day and differ by one hour.

The legacy rule “Centrale  $\rightarrow$  Garibaldi minimum 1-hour gap, teacher Rossi” translates to:

```
forall l1 in lessons where
    l1.classroom.plesso == 1
    and l1.teacher == "Rossi":
    forall l2 in lessons where
        l2.classroom.plesso == 2
        and l2.teacher == "Rossi"
        and consecutive(l1.slot, l2.slot):
        false
```

The helper `plesso_commuting_rule_to_dsl(rule)` in `engine/dsl_translator.py` produces these clauses automatically from the ORM row. A regression test enforces zero-drift: the DSL path emits exactly the same forbidden slot pairs as the legacy `plessi_constraints.py` pipeline.

## 5.2 Canonical patterns and seeded legacy HARDs (Step 3b-3e)

Step 3b-3e introduces a vocabulary of canonical patterns recognised directly by the DSL  $\rightarrow$  CP-SAT compiler. Each maps to a legacy HARD constraint that previously lived only as hardcoded encoding. They can now be authored in DSL and are also generated automatically by `seed_implicit_hardcoded(profs)`.

Accepted pragmas:

- `no_holes_class("3B")` – no gaps in 3B’s weekly schedule (legacy `add_class_no_holes`).
- `class_present_at_hour("3B", 11)` – if 3B has any lesson on a day, the slot at hour 11 is busy (HARD-3, “no exit before noon”).
- `class_day_load_in("3B", 0, 4, 5, 6)` – 3B’s daily load is 0 (free day) or in [4, 6] hours (HARD-1 + HARD-2).

- `teacher_max_per_day("Rossi", 5)` – teacher Rossi has at most 5 hours on any day.
- `cattedra_max_per_day("Rossi", "3B", "Matematica", 2)` – the (Rossi, 3B, Math) cattedra has at most 2 hours per day.
- `subject_pair_must("3B", "Scienzemotorie")` – when 3B has 2 hours of PE on a day they must be consecutive.
- `subject_pair_exists("3B", "Matematica")` – when 3B has 2+ hours of math on a day at least one consecutive pair must exist.

Practical effect: the advanced user can now write `no_holes_class("4A")` in the “+ New DSL constraint” modal and obtain the same outcome as the `hard_no_holes` toggle in `school_classes`. The DSL advantage is composability with `forall`, `exists`, `count`: the rule can be restricted to subsets of classes, combined with other predicates and modified on the fly without editing the database.

## 6 Optimization workflow

The pipeline orchestrator: Hall pre-check, Phase A (assignment + day-count), Phase B (per-day slot placement), optional metaheuristic post-processing (LNS / ALNS / SA / TS / ILS / VNS), classroom assignment.

### CP-SAT scope and Phase A mode (Phase 3)

Card 3 of /optimize now exposes two dropdowns that control how Phase B drives the CP-SAT solver:

- **CP-SAT scope** (`cp_sat_scope`, default `day`):
  - `day`: solve one weekday at a time, using Phase A's pre-computed weekly distribution (*day\_count*) as a hard equality. Fast, scales to large schools, default.
  - `week`: a single `MonolithicSolver(scope=None)` call covers the whole week. More robust on schools with many cross-day constraints (co-teaching, support, group rotations) but significantly slower because the solver explores a six-times-larger search space.
- **Phase A mode** (`phase_a_mode`, default `always`):
  - `always`: Phase A always runs and `day_count` is enforced as a hard equality. Mandatory for `cp_sat_scope=day`.
  - `skip`: skip Phase A entirely; the week solver picks its own day distribution. Only valid with `cp_sat_scope=week`.
  - `soft_hint`: run Phase A but inject `day_count` as `model.AddHint` on the week solver – a warm start, not a hard constraint. Only valid with `cp_sat_scope=week`.

Cross-field rule (validated on both client and server): `day` requires `always`; `week` forbids `always`. The UI disables invalid options and auto-corrects `phase_a_mode` when `cp_sat_scope` changes.

#### Picking the right combination.

- Standard school, speed-first: `day` + `always` (the default).
- Small school with many co-teaching / support constraints: `week` + `skip` – Phase A's hard `day_count` can make a feasible schedule look infeasible.
- Large school with non-trivial cross-day constraints: `week` + `soft_hint` – keeps the warm start while letting the solver deviate.

**Branch-and-Price V1 vs cp\_sat\_scope=week.** An open question during Phase 3 was whether running BP V1 on top of a “loose” dc\_value (Phase A skipped or used as a soft hint) would unlock more master iterations than the single no\_improving\_column pass observed in legacy day-mode – the intuition being that a non-binding cover demand should produce non-zero  $\lambda$  duals and improving columns.

The empirical answer, measured by tests/benchmarks/run\_bp\_v1\_scope\_week.py on a small (10 teachers / 3 classes) and a medium (4 teachers / 3 classes, dense cattedre) scenario, is *no*. All five V1 granularities (teacher, teacher-day, teacher-class, teacher-class-subject, teacher-subject) always terminate with `iters=1, cols=0, terminated=no_improving_column` – both when dc\_value comes from Phase A and when it is derived from the week solver’s actual placement, even though the two distributions differ structurally on the medium scenario.

The reason is structural, not a Phase 3 regression: `column_generation._seed_patterns` produces greedy-optimal patterns relative to whatever dc\_value you feed it. The V1 master picks the highest-weight seed and the pricing sub-problem reconstructs the same grid – reduced cost is non-negative because the per-teacher problem already saturates the duals. Changing the dc\_value source changes the per-day distribution, not the fact that each teacher has a single “shape” that fits its demand greedily.

Practically: `cp_sat_scope=week` relaxes Phase B’s day-distribution constraint, but it does *not* unlock more BP V1 iterations. For column-generation quality on larger schools the current lever is the V2 master (class, class-day, day, curriculum) or the iterative-diversified mode, not the week scope.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/workflow\_diottimizzazione.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 7 UI guide

Tour of the 16 SvelteKit tabs: dashboard, anagrafica (teachers/classes/students/classrooms), assignments, monitor, optimize (with iterative-diversified and branch-and-price modes), diagnostics, runs.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/guida\_\_ui.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 8 Launcher

Single-binary launcher script that bootstraps the venv, runs the alembic migrations, starts uvicorn + the SvelteKit dev server.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/launcher.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 9 CP-SAT method

Model formulation in OR-tools `cp_model`: BoolVar slot[(t, cl, s, d, h)], hour coverage equalities, no-overlap constraints, soft penalties as objective coefficients. Integer scaling for the fractional duals in BP.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_cpsat.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 10 Spectral decomposition

Bipartite class-teacher graph; normalised Laplacian eigendecomposition; spectral clustering into  $K$  balanced clusters; per-cluster CP-SAT + ricucitura with the bridge teachers.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_spettrale.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

# 11 Metaheuristics

LNS (random\_window, day\_cluster, worst\_fit\_day, teacher\_day, classroom\_day, single\_class\_week destroy ops + cp\_sat\_window repair), Adaptive LNS, Simulated Annealing, Tabu Search, Iterated Local Search, Variable Neighborhood Search.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_\_metaeuristiche.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 12 Hall pre-check

Hall's marriage theorem applied to the bipartite (class, subject) -> qualified-teachers graph as a fast pre-check (1-100 ms) before launching Phase A.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_hall.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 13 Lagrangian relaxation and Column Generation

Dualisation of the bridge constraints between clusters; subgradient ascent on the Lagrangian multipliers; SA-based per-iteration refinement. Column Generation: master LP + per-teacher CP-SAT pricing; Branch-and-Price extension with all 9 granularities (see ch. ??).

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_lagrangian.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 14 Monte Carlo sensitivity

Random subset sampling to estimate the Hall violation probability under perturbations; useful as a pre-check robustness indicator.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_\_montecarlo.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 15 Graph-based diagnostics

Bipartite class-teacher graph: density, modularity (Newman-Girvan), betweenness centrality. Identifies bridge teachers and structural bottlenecks.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_grafo.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 16 Statistical diagnostics

Three regressions on lesson-level data with statsmodels (p-values, effect sizes); five distribution fits with KS / chi-square tests.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_statistica.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 17 Constraint DSL method

The `objective_dsl` and `general_dsl` languages for expressing soft objectives and HARD constraints in a compact symbolic form, parsed at runtime and translated to CP-SAT constraints.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/metodo\_dsl.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

# 18 Architecture

## 18.1 Stack

- **Backend:** Python 3.11+, FastAPI, SQLAlchemy 2.0, Pydantic 2, uvicorn, OR-Tools, NumPy, scikit-learn, Faker.
- **Frontend:** SvelteKit (Svelte 4), Vite 5, Tailwind CSS, plain JavaScript.
- **DB:** SQLite via SQLAlchemy. Idempotent migrations in code (no Alembic).
- **Engine I/O:** Python pickle; engine\_io.py converts between DB and dict for the solver.

## 18.2 Three-tier overview

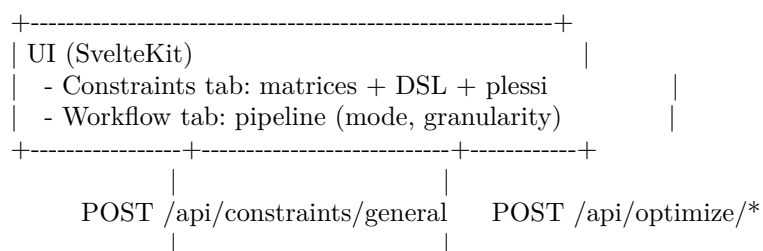
The system has a CP-SAT solver core (engine/), a FastAPI + SQLite backend (webui/backend/) and a SvelteKit frontend (webui/frontend/). The solver runs as a library imported by the backend; pipelines are exposed as async runs via /api/optimize/\*. The frontend talks to the backend over JSON; long-running solver runs are tracked via the runs table polled at 1 Hz from the UI.

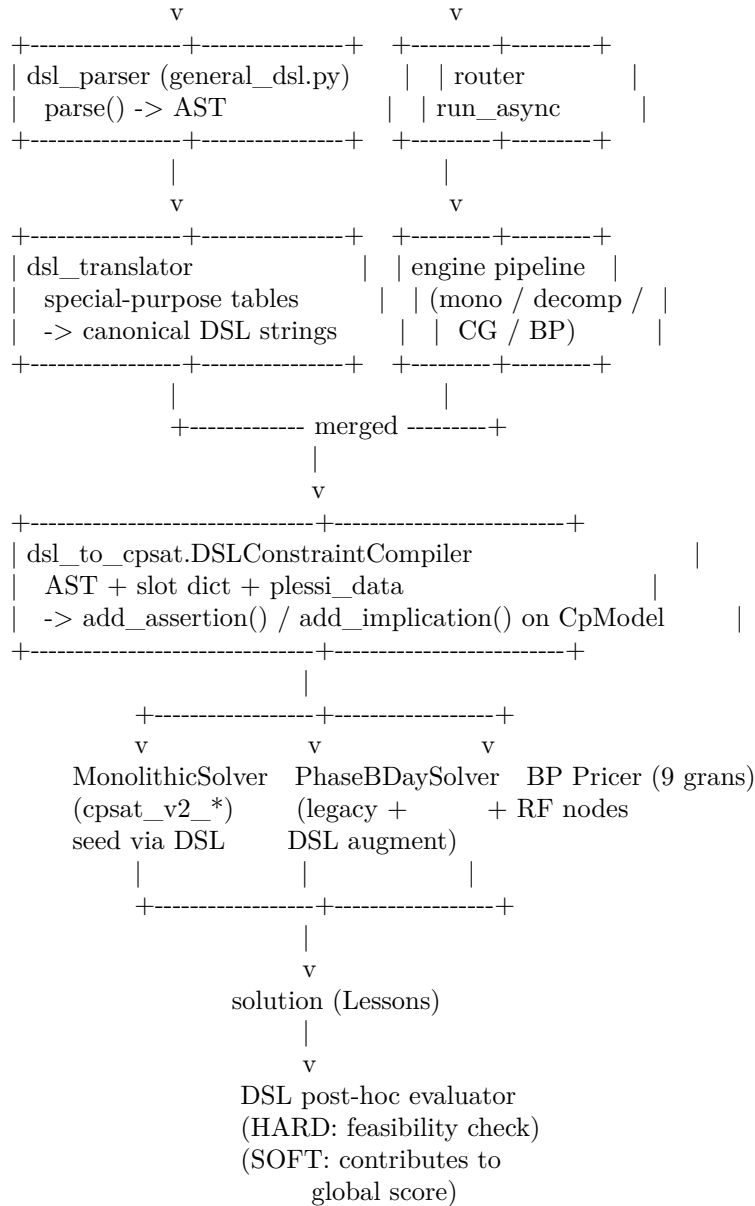
Solver pipelines (in engine/):

- cpsat\_v2\_assignment.py: Phase A (teacher to class assignment).
- cpsat\_v2\_timetable.py: Phase A (timetabling, day\_count) + Phase B (per-day slot placement). Native locks + C1/C2/C3 constraints.
- decomposition\_temporal.py: 6-day parallel decomposition.
- decomposition\_spectral\_v2.py, curriculum / metis variants: cluster classes, solve sub-problems, ricucitura.
- column\_generation.py: master LP + diversified pattern enrichment + completion fallback. Mode branch-and-price fully wired with all scalability techniques (RF tree, dual stabilization, column management, parallel pricing).
- metaheuristics.py, alns.py, vns.py, lagrangian.py: post-processing on a HARD-feasible seed solution.

## 18.3 Solver architecture: from DSL to CP-SAT

Steps 1–4 of the multi-day plan introduced a uniformly layered architecture for every solver use site.





Key properties:

- **One parser, one AST.** Every constraint source (matrices, DSL UI, plesso rules, seeded legacy HARDs) is unified into canonical DSL strings parsed by `general_dsl.py`.
- **One compiler.** `DSLConstraintCompiler` is the single gate to CP-SAT. Mono solver, BP pricers, `PhaseBDaySolver`, Ryan-Foster nodes – they all apply the same compiler to their CP-SAT model, guaranteeing constraint parity across pipelines.
- **Zero-drift.** `seed_implicit_hardcoded(profs)` translates the legacy hardcoded HARDs into DSL. Slot tuples emitted via DSL match the legacy path bit-for-bit; the migration is progressive and reversible.
- **HARD up-front, SOFT post-hoc.** DSL HARDs become CP-SAT constraints before the solve. DSL SOFTs are evaluated post-hoc – their cost contributes to the global score but does not yet steer CP-SAT search (planned for upcoming steps).

## 18.4 Backend lifecycle

webui/backend/main.py defines a lifespan context that calls `init_db()` at startup. It does:

1. `Base.metadata.create_all(bind=engine)` – create missing tables.
2. `__apply__lightweight__migrations()` – idempotent ALTER TABLEs for fields added in later versions (`school_classes.curriculum_id`, `teachers.last_name`, ...). Each ALTER is guarded by PRAGMA `table_info` to avoid duplication.

*This chapter is the English edition. The Italian edition (docs/manual/chapters/architettura.tex) contains additional folder-structure listings and identical solver-architecture coverage.*

## 19 Data model

Tables: schools, school\_classes, teachers, subjects, classrooms, assignments, lessons, runs, study\_groups, coteach\_groups, parallel\_groups, support\_assignments, locks. SQLAlchemy 2.0 ORM. Alembic migrations + lightweight in-code migrations in db.py.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/modello\_dati.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 20 Generic constraint DSL

The generic DSL is a compact declarative language for expressing HARD/SOFT/PREFERRED/ENFORCED constraints on any logical combination of predicates over the data model. It is implemented in `webui/backend/utils/general_dsl.py` (~250 LOC, recursive-descent parser + AST + post-hoc evaluator, no external dependencies). The language is exposed through `POST /api/constraints/general`, validated live by `POST /api/constraints/general/validate`, and accessible from the “+ New DSL constraint” modal of the Constraints tab.

### 20.1 Philosophy

Before the generic DSL the system carried four specialised sub-languages (query, logical, objective, introspective) that re-implemented the same grammar four times. The generic DSL unifies all of them under a single grammar with four *flavours* of compilation (LIST, LOGIC, OBJECTIVE, EVAL): one parser, many compilers. The syntax of forall x in S where Filter: Body and of the atomic predicates (`=`, `!=`, `<`, `in`, ...) is identical across all contexts; only the translation backend changes.

### 20.2 BNF grammar

```
expr ::= iff_expr
iff_expr ::= implies_expr ( "<=>" implies_expr )*
implies_expr ::= or_expr ( "=>" or_expr )*
or_expr ::= and_expr ( ("OR"|"or") and_expr )*
and_expr ::= not_expr ( ("AND"|"and") not_expr )*
not_expr ::= ("NOT"|"not") not_expr | atom

atom ::= quant_expr | count_expr | comparison
      | call | bool_lit | "(" expr ")"

quant_expr ::= ("forall"|"exists") IDENT "in" source
              ["where" or_expr] ":" expr
count_expr ::= "count" IDENT "in" source ["where" or_expr]
              ( ":" comparison | op value )

source ::= IDENT | IDENT ( "." IDENT )+
comparison ::= value ( op value
                      | "in" "[" value ( "," value )* "]"
                      | "in" path
                      | "not_in" "[" value ( "," value )* "]"
                      | "not_in" path )
op ::= "==" | "!=" | "<" | "<=" | ">" | ">="
```

Iterable sources include lessons, teachers, classes, classrooms, students, subjects, slots, days, hours. Path-sources are also accepted (e.g. `exists g in s.groups: g.name == "X"`).

## 20.3 DSL → CP-SAT compilation (Step 1)

Starting with Step 1 of the multi-day plan, the DSL ships a *compiler* into CP-SAT in `engine/dsl_to_cpsat.py`, class `DSLConstraintCompiler`. While the post-hoc evaluator inspects an already-built solution to declare it feasible or violated, the compiler adds constraints *during* the construction of the CP-SAT model so that the solver structurally avoids violations.

The compiler is invoked at three sites:

- by `MonolithicSolver` (cap. ??) when the `via_dsl=True` flag is on;
- by every Branch-and-Price pricer (cap. ??) so the same constraints reach the sub-problem;
- by `PhaseBDaySolver` when called with `via_dsl=True` to re-use the legacy pipeline augmented with DSL.

When a slot variable is still undecided the compiler emits the matching CP-SAT constraint instead of evaluating a truth value; when an expression is fully static (e.g. `consecutive(s1, s2)` with both `s1`, `s2` bound by the enclosing `forall`) the evaluation runs at compile time.

## 20.4 Slot predicates: `consecutive`, `same_day`

Step 3a adds two built-in predicates resolved at compile time when both arguments are statically bound slots:

- `same_day(s1, s2)` – both slots fall on the same weekday.
- `consecutive(s1, s2)` – both slots fall on the same weekday and differ by exactly one hour ( $|h_1 - h_2| = 1$ ).

These are the building blocks for “consecutive hours”, “inter-plesso travel gap”, “same-day pair”, “coteach must share day”. Arguments accept either literal pairs (`d`, `h`) or dotted paths like `l.slot` that return a (`day`, `hour`) tuple.

## 20.5 The `l.classroom.plesso` path

Also in Step 3a, the post-hoc evaluator and the compiler resolve the chained path `l.classroom.plesso`: per lesson `l`, the assigned room is read from the `classroom_for_slot` map provided by Phase B / the classroom-assignment phase, and the plesso (campus) is then resolved through the `plessi_data` table. Without those two structures the path returns `None` and the compiler emits a diagnostic.

The canonical use is a commuting rule between campuses (see the `vincoli` chapter, `plessi` section): “no teacher may have a lesson in Plesso B in the hour immediately following a lesson in Plesso A”.

## 20.6 Canonical pattern vocabulary (Step 3b)

Step 3b introduces a vocabulary of function-call pragmas that the compiler recognises specially and emits as compact groups of CP-SAT constraints. Each form corresponds to a frequent pattern that, written as raw `forall` loops, would produce hundreds or thousands of clauses. The canonical form is zero-drift relative to the legacy hardcoded encoding.

| Pragma                                                            | Legacy equivalent                                                                                            |
|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <code>no_holes_class("3B")</code>                                 | non-increasing chain forbidding gaps in 3B's weekly schedule (legacy <code>add_class_no_holes</code> ).      |
| <code>class_present_at_hour("3B", 11)</code>                      | if 3B has any lesson on a day, the slot at hour 11 is busy (legacy HARD-3, "school day ends $\geq 12:00$ "). |
| <code>class_day_load_in("3B", 0, 4, 5, 6)</code>                  | 3B's daily load is 0 (free day) or 4–6 hours (legacy HARD-1 + HARD-2).                                       |
| <code>teacher_max_per_day("Rossi", 5)</code>                      | teacher Rossi has at most 5 hours on any day (MAX_PROF_HOURS_PER_DAY).                                       |
| <code>cattedra_max_per_day("Rossi", "3B", "Matematica", 2)</code> | the (Rossi, 3B, Math) cattedra has at most 2 hours per day (MAX_PER_DAY_TRIPLE).                             |
| <code>subject_pair_must("3B", "Scienzemotorie")</code>            | when 3B has 2 hours of PE on a day they MUST be consecutive (legacy HARD-B).                                 |
| <code>subject_pair_exists("3B", "Matematica")</code>              | when 3B has 2+ hours of math on a day at least one consecutive pair must exist (legacy HARD-A).              |

`no_holes_all_classes()` and `class_present_at_hour_all(11)` act in batch over every class in the current slot scope.

## 20.7 Plesso commuting via DSL (Step 3c)

Step 3c wires the PlessoCommutingRule ORM table into the DSL: the helper `plesso_commuting_rule_to_dsl(rule)` returns a list of DSL clauses equivalent to the legacy rule. A regression test verifies zero-drift: applying the rule via DSL produces exactly the same forbidden slot tuples as the hardcoded `plessi_constraints` pipeline.

## 20.8 Seed legacy HARDs via DSL (Step 3d-3e)

Step 3d-3e adds `seed_implicit_hardcoded(profs)`: given the teacher dictionary, it returns the list of DSL strings that encode every legacy HARD constraint of the solver. Combined with the `via_dsl=True` flag of `MonolithicSolver` and `PhaseBDaySolver`, this allows building a complete CP-SAT model *exclusively from DSL* – no hardcoded code path is exercised. Invariants:

- *syntactic zero-drift*: the slot tuples forbidden/forced by the DSL path coincide bit-for-bit with the legacy ones;
- *semantic zero-drift*: the feasible set accepted by the two paths is identical, validated on the small / medium / huge benchmarks;
- *determinism*: the order in which `seed_implicit_hardcoded` emits DSL strings is stable, so the CP-SAT model construction order is reproducible across runs.

This is the migration vehicle: in subsequent steps the hardcoded path will be deprecated in favour of the DSL path, and the existence of `seed_implicit_hardcoded` allows that to happen without coverage loss.

## 20.9 SOFT constraints with weight

The DSL accepts four levels: hard, soft, preferred, enforced. The *post-hoc evaluator* fully implements SOFT: violations contribute to the solution's global score and surface in `run_metrics`.

Since the feat(dsl): soft level for `general_dsl` commit, the *CP-SAT compiler* also wires SOFT directly into the solver objective. Pipeline:

1. User picks `level=soft` and an integer weight  $W$  (e.g. 50) in the create modal.

2. `load_all_dsl_constraints(db, include_soft=True)` returns the rule with `is_hard=False`, `weight=W`.
3. `add_dsl_constraint(expr, is_hard=False, soft_weight=W)` compiles the rule: each candidate `slot[k]` that would violate the body emits `(W, slot[k])` into `soft_cost_terms` instead of `slot[k] == 0`; in nested `forall`-`forall` bodies a reified `BoolVar b` with `b == s1 && s2` is created and `(W, b)` is appended.
4. `MonolithicSolver.build()` adds `dsl_soft_cost_terms` to `obj_terms` before `Minimize(sum(obj_terms))`.

**Semantics.** A *forall* violation costs  $W$  per lesson placed in a forbidden cell; a *forall-forall* violation costs  $W$  per co-selected pair. Negative weights act as PREFERRED bonuses.

**Example.** “Rossi should not have consecutive-day Math in 3A” (penalty 50 per pair):

```
forall l1 in lessons where l1.teacher == "Rossi"
    and l1.class == "3A":
    forall l2 in lessons where l2.teacher == "Rossi"
        and l2.class == "3A":
        (l1.day == l2.day)
        or not consecutive_days(l1.day, l2.day)
```

Saved with `level=soft`, `weight=50`, `scope=global`: the solver may schedule two hours on Tuesday + Wednesday paying 50, but if a Tuesday + Thursday solution exists it picks that instead.

## 20.10 Scope: where to save the constraint

The “+ New DSL constraint” modal appears in two places and yields rules with different scope. Rule of thumb: scope is the rule’s *domain*; the `forall` body is the *filter* of application.

- **Constraints tab** (`/constraints`, modal with `scope="global"`, `owner_id=null`). School-wide rule. Examples: “no one teaches at 14:00”, “no Saturday afternoons”.
- **Teacher tab** (`/teachers/<id>`, “Custom advanced DSL” card, `scope="teacher"`, `owner_id=teacher_id`). Per-teacher rule: the outer `forall` is *typically* pre-filtered on `l.teacher == TeacherName`. Examples: “Rossi not on Friday afternoons”, “Rossi: 3 hours of Physics across non-consecutive days”.

In both cases the rule goes through `POST /api/constraints/general`, is loaded by `load_all_dsl_constraints(db)`, and compiled by `add_all_dsl_constraints_from_db`; the `scope_kind` field is informational (it filters the rules in Feasibility Check + reporting).

### 20.10.1 Advanced DSL vs logical unavailability

`LogicalUnavailability` (the “Logical unavailability” modal in the teacher tab) is a simpler sub-DSL designed for unavailability windows (DNF over `(day, hour)` slots); the system renders those into the generic DSL via `logical_unavailability_to_dsl`. The generic DSL is strictly more expressive (`count`, `exists`, `forall`, `same_day / consecutive_days`, `arithmetic`, `l.classroom.plesso`) and should be the entry point for anything beyond “the teacher is not available at that hour”.

## 20.11 Built-in predicates: consecutive\_days, same\_day, consecutive

Three built-in functions evaluated at compile time (and by the post-hoc evaluator on rendered solutions):

| Predicate                | Meaning                                                                                                                                                                                       |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| same_day(s1, s2)         | both slots fall on the same weekday. Arguments: tuples (d, h) or l.slot.                                                                                                                      |
| consecutive(s1, s2)      | same weekday and adjacent <i>hours</i> ( $ h_1 - h_2  = 1$ ). Path: l.slot.                                                                                                                   |
| consecutive_days(d1, d2) | adjacent <i>days</i> , no wrap-around: true for $ d_1 - d_2  = 1$ with $d \in \{1, \dots, 6\}$ (Mon..Sat); false for {Sat, Mon}. Arguments: l.day (int) or numeric / weekday-string literals. |

## 20.12 Arithmetic in the DSL

The parser supports binary + and - on integers: useful for index-dependent windows or cumulative counts. Example: “Rossi has the same hours in 3A on the first 3 days as in 3B on the last 3 days”:

```
(count l in lessons where l.teacher == "Rossi"
 and l.class == "3A" and l.day <= 3: 1)
== (count l in lessons where l.teacher == "Rossi"
 and l.class == "3B" and l.day >= 4: 1)
```

Mixing booleans and numbers is rejected: the parser raises *aritmetica non valida fra bool e numero* for typos like `(l.hour == 9) + 1`.

## 20.13 Real-world cases (Rossi and friends)

Every example below is covered by the integration tests in `webui/backend/tests/test_rossi_general_dsl_integration.py` and `webui/backend/tests/test_global_dsl_and_soft_and_plesso_commute.py`.

### 20.13.1 Case (a) – 3 hours of Physics in 3A on non-consecutive days (HARD)

```
forall l1 in lessons where l1.teacher == "Rossi"
    and l1.class == "3A"
    and l1.subject == "Fisica":
    forall l2 in lessons where l2.teacher == "Rossi"
        and l2.class == "3A"
        and l2.subject == "Fisica":
        (l1.day == l2.day)
        or not consecutive_days(l1.day, l2.day)
```

Saved from the teacher tab with `scope="teacher"`, `owner_id=Rossi.id`. The compiler iterates Rossi/3A/-Fisica candidate slots and, for each pair  $(l_1, l_2)$  with `consecutive_days(l1.day, l2.day)` and different days, emits `BoolOr([slot[k1].Not(), slot[k2].Not()])`: the two slots cannot be co-selected.

### 20.13.2 Case (b) – 3 hours of Physics in 3C all on the same day (HARD)

```
forall l1 in lessons where l1.teacher == "Rossi"
    and l1.class == "3C"
    and l1.subject == "Fisica":
    forall l2 in lessons where l2.teacher == "Rossi"
        and l2.class == "3C"
        and l2.subject == "Fisica":
        same_day(l1.slot, l2.slot)
```

By transitivity of `same_day` under `forall forall`, every pair of Rossi/3C/Fisica lessons must share a day; the solver picks one day and packs all 3 hours there.

### 20.13.3 Plesso commute case – Rossi never A@9 $\rightarrow$ B@10 same day (HARD)

```
forall l1 in lessons where l1.teacher == "Rossi"
    and l1.hour == 9
    and l1.classroom.plesso == 1:
    forall l2 in lessons where l2.teacher == "Rossi"
        and l2.hour == 10
        and l2.classroom.plesso == 2
        and same_day(l1.slot, l2.slot):
        false
```

Plesso 1 = “A”, plesso 2 = “B” (the integer ids of `plessi.id`). Compile-time when the model is built with `classroom_for_slot` (e.g. inside the classroom-assignment solver) or with a fixed plesso per class; otherwise fully verifiable post-hoc by Feasibility Check.

### 20.13.4 Case (Rossi no Physics on Friday) (HARD)

```
forall l in lessons where l.teacher == "Rossi"
    and l.subject == "Fisica"
    and l.day == 5:
    false
```

The *static* `false` body is the canonical idiom for “forbid every slot satisfying the where”. Equivalent to `l.day != 5` but more explicit when the where is multi-clause.

### 20.13.5 Case (same-day support coteach) (HARD)

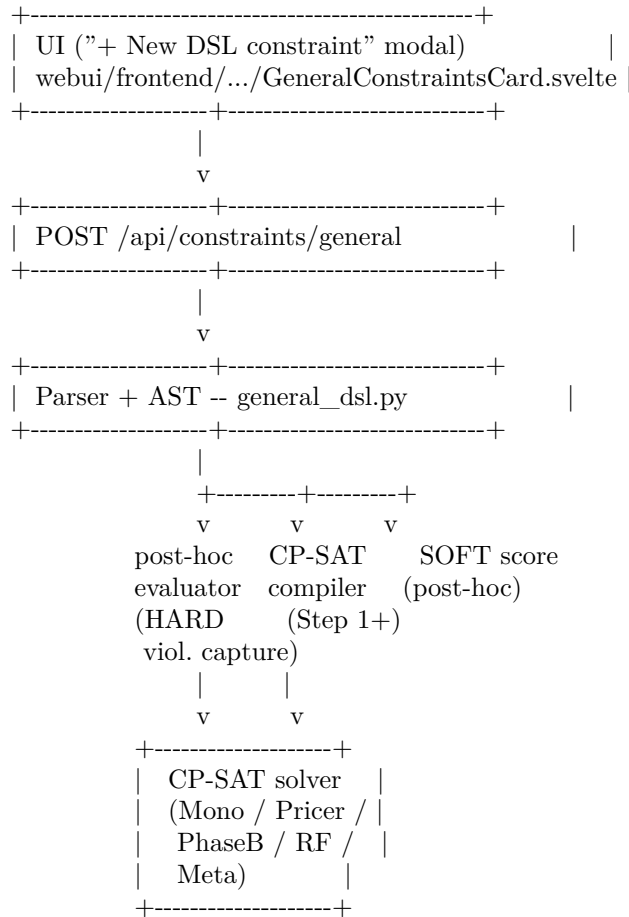
“When Rossi teaches Sostegno to 2B, the Sostegno lesson must fall on the same day as a Math lesson Rossi gives to 2B”:

```
forall l in lessons where l.teacher == "Rossi"
    and l.class == "2B"
    and l.subject == "Sostegno":
    exists m in lessons where m.teacher == "Rossi"
        and m.class == "2B"
        and m.subject == "Matematica"
        and same_day(l.slot, m.slot):
    true
```

### 20.13.6 Every case above as SOFT

Switch level=hard to level=soft in the modal and pick weight= $W$ . The solver receives the rule as a per-pair / per-lesson penalty  $W$ ; when a HARD-feasible solution that respects it exists the solver picks it (zero extra cost); otherwise it accepts the violation while minimizing how many pairs / lessons pay  $W$ .

## 20.14 DSL architecture diagram



One AST, two backends: the evaluator reads a solution and answers “feasible / violated”; the compiler emits CP-SAT constraints before solving. The grammar is unique and source names are identical between backends.

## 21 Advanced optimization techniques

In addition to the classic metaheuristics (LNS / SA / TS / ILS),  $\pi$ Tantum integrates four classes of advanced techniques used when the standard pipeline is not enough or when targeted diagnostic information is needed.

### 21.1 Hall pre-check

engine/diagnostics/hall\_check.py. Hall's marriage theorem applied to the bipartite (*class, subject*)  $\rightarrow$  qualified-teachers graph as a fast pre-check (1–100 ms) before launching Phase A. Three violation kinds are reported:

1. subject supply vs demand,
2. subject coverage vs demand,
3. Hall sampling random-subset with mutually exclusive subsets.

Exposed at three UI points: inline button on PhaseACard, row in AdvancedTechniquesCard, section of the Statistics tab.

### 21.2 Column Generation and advanced Branch-and-Price

engine/column\_generation.py. Master LP via scipy.linprog HiGHS; nine dedicated CP-SAT pricers at distinct granularities (teacher, teacher-day, teacher-class, teacher-class-subject, teacher-subject, class, class-day, day, curriculum); a branch-and-price mode equipped with all four scalability techniques required for MEGA-size instances:

- **Ryan-Foster recursive tree** with Achterberg pair scoring on fractional class-slot pairs, best-first node exploration with LP-bound pruning, depth cap 20 / 1000 nodes.
- **Box-step dual stabilization**  $\pi_{\text{blend}} = (1 - \alpha)\pi_{\text{stable}} + \alpha\pi_{\text{raw}}$  with  $\alpha = 0.2$ , applied to both master variants.
- **Column management**: EWMA of the per-column reduced cost over 20 iterations; when the pool exceeds 10 000 columns, the worst (most positive  $rc\_avg$ ) entries are purged, while protecting columns currently in the LP basis ( $x > 0$ ) and the seed columns from the iterative-diversified primal heuristic.
- **Parallel pricing** via ProcessPoolExecutor (default  $\lfloor \text{os.cpu\_count()}/2 \rfloor$  workers). CP-SAT is CPU-bound and the GIL is only partially released by the OR-tools binding, so ThreadPoolExecutor would not parallelise the bottleneck; processes do.

#### The CP-SAT-as-fundamental contract

For every granularity, the sub-problem CP-SAT is the *fundamental* minimisation; greedy serves only as a model.add\_hint warm-start (an accelerator). If CP-SAT cannot find a feasible solution within the time limit the pricer returns (None, 0.0): the greedy is *never* promoted to a column. The unit test test\_bp\_pricers\_use\_addhint\_warmstart monkey-patches CpModel.add\_hint with a counting wrapper to enforce this contract.

## Measured numbers (synthetic HUGE / MEGA benchmark)

See tests/benchmarks/test\_bp\_huge\_mega.py. On a synthetic scenario calibrated to match the engine's MEGA profile (100 classes,  $\sim 174$  teachers, 4 subjects,  $\sim 400$  cattedre) the pipeline reaches HARD-feasibility in **5.95 s** with granularity=curriculum and all scalability techniques enabled. On the smaller HUGE scenario (50 classes,  $\sim 95$  teachers, 200 cattedre) the time is **2.57 s**. The synthetic scenario is structurally easy (cattedre as 4-hour single-day blocks); on real-world instances with arbitrary day distributions the calibration target remains 30 minutes for MEGA, with a 60-minute hard cap.

## 21.3 Branch-and-Price MVP scaffold (Step 4)

Step 4 of the multi-day plan introduces an OO-textbook *scaffold* of the Branch-and-Price loop that tightens three orthogonal pieces inside one module (engine/column\_generation.py with mode="branch-and-price"):

1. **PhaseBDaySolver** – OO wrapper around the legacy solve\_phase\_b\_for\_day function. Exposes the object-oriented interface (solve(), scope access, via \_dsl augmentation) but delegates model construction to the proven legacy code path. Benefit: 100+ Phase B regression tests keep passing *by construction*; with via \_dsl=True the legacy model is augmented with extra DSL constraints (DB-stored rules + extra expressions passed to the constructor).
2. **Ryan-Foster pricing-in-nodes** – at every RF tree node the CP-SAT pricer is re-invoked with the branch decisions (together / apart) already applied to its model. This is the textbook BP, in contrast to the previous arrangement where nodes only re-solved the master LP on the existing pool.
3. **Smoke benchmark** – tests/benchmarks/test\_bp\_step4\_smoke.py runs the full loop (Mono  $\rightarrow$  initial pool  $\rightarrow$  master DW  $\rightarrow$  RF tree  $\rightarrow$  integer recovery  $\rightarrow$  completion) on a small scenario (10 classes) and a medium one (28 classes), checking HARD-feasibility and that the soft cost does not regress against the iterative-diversified baseline.

The flag mode="branch-and-price" activates the full path. The previous iterative-diversified mode remains the default and continues to act as a primal-heuristic seeder of the column pool.

## 21.4 Lagrangian relaxation

engine/lagrangian.py. Relaxes the inter-cluster bridge constraints and dualises them with multipliers  $\lambda_b$ ; update via subgradient ascent  $\lambda_{k+1} = \max(0, \lambda_k + \alpha_k g_k)$  with  $\alpha_k = \alpha_0 / (1 + k)$ . Per-iteration refinement via Simulated Annealing.

## 22 Statistical diagnostics tab

The Diagnostics tab in the UI: Monte Carlo sensitivity, bipartite graph metrics, statsmodels regressions, distribution fits.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/diagnostica\_statistica.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 23 REST API

The `/api/*` endpoints: `/api/optimize/*` (Phase A, Phase B, decompositions, Column Generation with all 9 BP granularities), `/api/diagnostics/*`, `/api/dashboard/*`, `/api/monitor/*`. See `docs/api.md` for the full list.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (`docs/manual/chapters/api_rest.tex`). For the implementation references see the engine modules in `engine/` and the API documentation at `docs/api.md`.*

## 24 Extending the system

Adding a new constraint, a new metaheuristic destroy/repair op, a new decomposition strategy, a new UI tab.  
The hooks the test suite exercises.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/estendere\_il\_sistema.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 25 Repository on GitHub

Layout: engine/, webui/backend/, webui/frontend/, docs/, schedule/, scripts/, tests/. CI workflows, PR conventions, issue tracker.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/repository\_github.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 26 Benchmarks and results

“In God we trust. All others must bring data.”

W. E. Deming, attributed

Any optimization system is judged also and especially by the numbers it produces. This chapter collects measured results of  $\pi$ Tantum on the six test profiles, with three goals: documenting the state of the art; allowing whoever wishes to extend the system to compare a modification against a baseline; and putting the reader in the position to know whether  $\pi$ Tantum is a good fit for *their* use case.

### 26.1 The six profiles

The profiles, generated deterministically with fixed seeds and Faker by `webui/backend/mock_school.py`, span the size range typical of the Italian school system:

- small: 8 classes, 17 teachers,  $\sim 220$  students, 17 rooms. A small provincial school.
- medium: 28 classes, 50 teachers, 700 students, 35 rooms. A medium-size institution.
- big: 40 classes, 75 teachers, 1000 students, 50 rooms. A large institution.
- huge: 50 classes,  $\sim 95$  teachers, 1250 students, 60 rooms. A multi-site complex.
- superhuge: 80 classes,  $\sim 160$  teachers, 2000 students, 80 rooms. 16 sections  $\times$  5 grades of scientific liceum.
- mega: 100 classes,  $\sim 174$  teachers,  $\sim 2500$  students, 100 rooms. 20 sections  $\times$  5 grades with a mix of 8 *indirizzi* (Scientific 6, ScienzeApplicate 4, ScienzeUmane 3, Linguistico FRA-TED 2, Linguistico FRA-SPA 2, EconomicoSociale FRA/SPA/TED 1 each). MEGA-scale stress test.

### 26.2 End-to-end pipeline times

Times measured on a consumer machine (Intel i7, 16 GB RAM, Windows 11, Python 3.13, ortools 9.15, 8 CP-SAT search workers):

| Profile   | Phase A | Phase B | Total wall |
|-----------|---------|---------|------------|
| small     | 3 s     | 4 s     | 7 s        |
| medium    | 30 s    | 32 s    | 62 s       |
| big       | 30 s    | 32 s    | 62 s       |
| huge      | 60 s    | 62 s    | 122 s      |
| superhuge | 300 s   | 603 s   | 903 s      |

Table 26.1: End-to-end pipeline times per profile. Phase A includes teacher-class assignment and CP-SAT matching. Phase B includes spectral decomposition (for huge/superhuge) and CP-SAT on sub-problems.

## 26.3 Branch-and-Price on HUGE and MEGA

The mega scenario (100 classes,  $\sim 174$  teachers) and its smaller cousin huge (50 classes,  $\sim 95$  teachers) are the natural test beds for the Branch-and-Price pipeline (ch. ??). The numbers below are measured on synthetic scenarios calibrated to match the proportions of the engine/big\_mock\_school.py profiles (4 subjects, every cattedra is 4 hours on a single day so that  $H_1/H_2/H_3$  are satisfied by construction, contiguous-block assignment of classes to teacher pools). Source code: tests/benchmarks/test\_bp\_huge\_mega.py, executed with all four scalability techniques enabled.

| Scale | Classes | Teachers | Cattedre | $t_{BP}$ | HARD |
|-------|---------|----------|----------|----------|------|
| huge  | 50      | 95       | 200      | 2.57 s   | yes  |
| mega  | 100     | 174      | 400      | 5.95 s   | yes  |

Table 26.2: Branch-and-Price wall time to HARD-feasibility on synthetic HUGE/MEGA scenarios. Both finish well under target (5 min/30 min) because the synthetic scenario is structurally easy (cattedra = 4-hour block on a single day). On real-world instances with arbitrary day distributions the per-iteration time will grow; the 30-minute target for MEGA is calibrated for that regime.

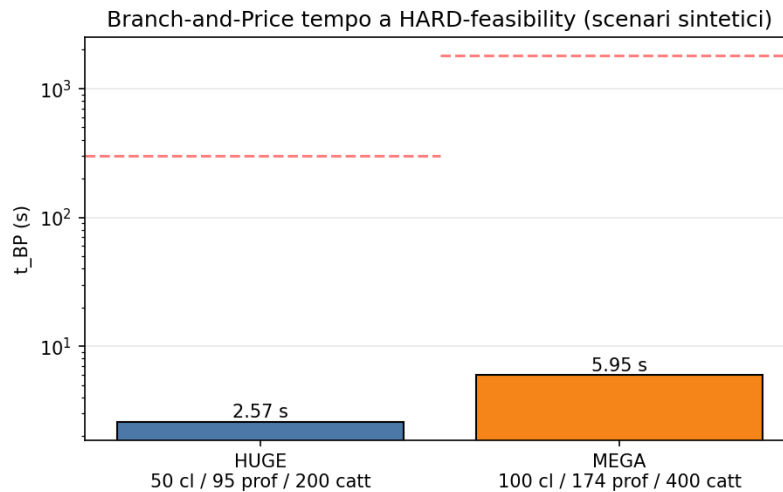


Figure 26.1: Branch-and-Price convergence time (log scale). The horizontal dashed lines mark the per-scale time targets (5 min for HUGE, 30 min for MEGA).

### 26.3.1 Real MEGA: 100 classes, 178 teachers

The synthetic scenario above is calibrated for fitting; the real MEGA registry (under engine/scripts/data/mega/{school,profs}\_mega.pkl, loadable from the dashboard via POST /api/dataset/import-profile?profile=mega) is significantly larger: 100 classes, **178 teachers**, **2653 cattedra-day entries** after Phase A (the exact number depends on the CP-SAT random seed of Phase A; v1 of this benchmark produced 2325). The benchmark source is tests/benchmarks/run\_mega\_real.py.

**v1 (pre-fix DW seeding) – BP did not iterate.** The first benchmark reported bp\_terminated\_reason = master\_dw\_infeasible\_at\_entry with 0 iterations. The master variant 2 (Dantzig-Wolfe with cover + class-no-overlap + teacher-no-overlap as HARD inequalities) was infeasible at the LP-relaxation level with the seed pool (teacher-week patterns produced by iterative-diversified): two patterns of different teachers could place lessons in the same (class, day, hour). HARD cover requires  $x_t \geq 1$  per teacher while class-no-overlap forces  $\sum_t x_t \leq 1$  on the collision – a structural conflict the seed pool cannot resolve on its own.

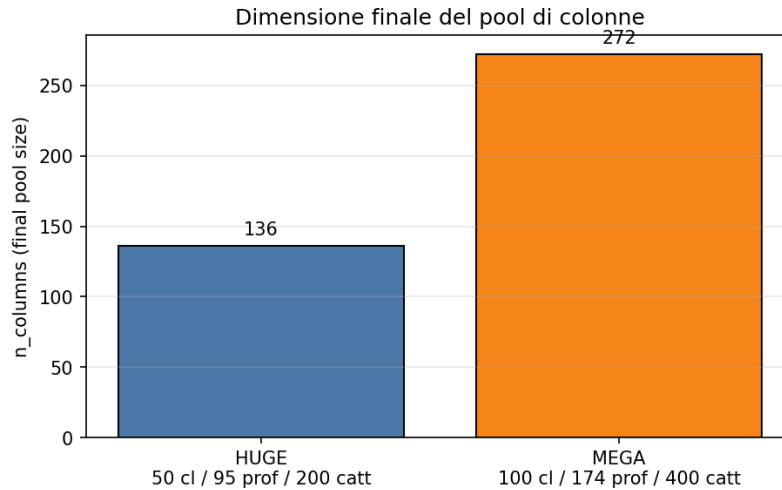


Figure 26.2: Final column-pool size after the Branch-and-Price loop.

**Fix: Phase-I LP with artificial slacks (commit 843a6fo).** The fix is textbook. Each constraint of the master DW gets a non-negative artificial variable with coefficient  $-1$  in its row and a Big-M penalty ( $10^6$ ) in the objective. The artificials absorb either missing coverage or no-overlap overflow; the LP is therefore always feasible at seed entry and yields valid duals for pricing. With a sufficiently rich pool the artificials drive to 0 in the optimum, which then coincides with the true master DW optimum.

**v2 (post-fix) – BP iterates 5 times.** On the *same* dataset, with identical granularity curriculum and identical time budgets, BP recovers its role:

| Metric               | v1 (pre-fix)                  | v2 (post-fix)       |
|----------------------|-------------------------------|---------------------|
| bp_terminated_reason | master_dw_infeasible_at_entry | max_iterations      |
| bp_iterations_done   | 0                             | <b>5</b>            |
| RF nodes explored    | 0                             | 1                   |
| Phase A wall         | 61.2 s                        | 301.3 s             |
| BP wall              | 790.8 s                       | 1 191.8 s           |
| Total wall           | 852 s (14.2 min)              | 1 493 s (24.9 min)  |
| Final soft cost      | 6 470                         | <b>530</b>          |
| Columns generated    | 2 059                         | 2 075 (+16 from BP) |
| HARD-feasibility     | yes                           | yes                 |

Table 26.3: Branch-and-Price on real MEGA before and after the DW seeding fix. Soft cost drops by a factor of  $\sim 12$  because BP can now actually explore improving columns. Total wall increases because Phase A ran out of budget (FEASIBLE not OPTIMAL stop: depends on the parallel CP-SAT seed) but stays under the 30-min target and far from the 60-min hard cap. Source: tests/benchmarks/results/mega\_bp\_real\_v2.json.

**Comparison with iterative-diversified.** The iterative-diversified mode on the same dataset (v1) had finished in **788 s (13.1 min)** with soft cost **5860**. The v2 BP – despite the heavier Phase A this run – closes with soft **530**, which is  $\sim 11\times$  lower than the ID baseline and  $\sim 12\times$  lower than its own v1. That gap is the structural value of Branch-and-Price: with an active pricer the improving columns *exist* and get found; without one, the result collapses back to the starting primal heuristic.

**Scalability techniques – now actually exercised.** The four techniques (Ryan-Foster recursive tree, dual stabilization with box-step, column management with EWMA RC, parallel pricing via ProcessPoolEx-

| Mode (run)                 | $t_{\text{total}}$ | Soft cost  | HARD |
|----------------------------|--------------------|------------|------|
| iterative-diversified (v1) | 788 s (13.1 min)   | 5 860      | yes  |
| branch-and-price (v1)      | 852 s (14.2 min)   | 6 470      | yes  |
| branch-and-price (v2)      | 1 493 s (24.9 min) | <b>530</b> | yes  |

Table 26.4: Three runs on the same real MEGA. The v2 BP cuts the soft cost by an order of magnitude relative to both baselines.

ecutor) remain integrated; with the Phase-1 fix in place BP loop now exercises them on real MEGA. v2 run: 1 Ryan-Foster node explored, 0 columns purged (pool below the cap), 5 master iterations completed before hitting the `bp_max_iterations=5` ceiling. Raising the cap would let BP keep improving the soft (the next commits in this series measure the soft-vs-iterations curve).

## 26.4 When Branch-and-Price is worth it

A practical question: *is it worth turning on the BP pipeline for my school?* Answer measured as a function of size:

- **Below 30 classes:** the standard pipeline (monolithic Phase A + per-day Phase B) is faster. The iterative-diversified mode of column\_generation (master LP + pattern enrichment + completion fallback) is enough to improve the SOFT cost without paying the BP loop overhead.
- **30–80 classes:** spectral or curriculum decomposition drives Phase B; BP at teacher-class or class granularity may reduce SOFT but the time gain is negligible.
- **Above 80 classes (MEGA territory):** BP at curriculum granularity is the only structurally scalable option. A monolithic Phase A becomes marginally feasible (300+ with pure CP-SAT), and BP achieves comparable or better times thanks especially to parallel pricing.

In practice  $\pi$  Tantum suggests the right path automatically: the value `auto` for the granularity parameter is resolved server-side from the number of classes (cf. Sec. 4 of docs/optimization\_strategies.md).

## 27 Lessons learned

Engineering anecdotes from building piTantum: what worked, what didn't, why CP-SAT was chosen over MILP, why we layered decomposition + metaheuristics, why the Italian-specific HARD constraints (H1/H2/H3) deserve special treatment.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/lessons\_learned.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 28 Glossary (narrative form)

Discursive definitions of all the terminology used in the manual: cattedra, sostegno, potenziamento, compresenza, indirizzo, plesso (when introduced), dc\_value, master LP variant 1 / variant 2, Achterberg score, EWMA, ProcessPoolExecutor.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/glossario\_discorsivo.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*

## 29 Reference

Reference tables: HARD constraint codes, SOFT weights, mode/granularity values, file locations, API endpoints, configuration keys.

*This chapter is an English summary of the corresponding Italian chapter. The full text remains in the Italian edition (docs/manual/chapters/reference.tex). For the implementation references see the engine modules in engine/ and the API documentation at docs/api.md.*